NYU TANDON SCHOOL OF ENGINEERING

NYU TANDON CAREER HUB · FALL 2026

Data Science Bootcamp

Week 6 · Machine Learning Fundamentals, Perceptron



Instructor: **Angelin Ann Jacob**

Section 1 · Tues Nov 3

NYU TANDON SCHOOL OF ENGINEERING

Useful Links

Zoom

All sessions are virtual. Use ONLY **your NYU SSO Login to be considered for attendance**

Brightspace

Assignments, submissions and the discussion board, monitored Monday to Thursday.

GitHub

Slides and Colab notebooks for every session.

Email

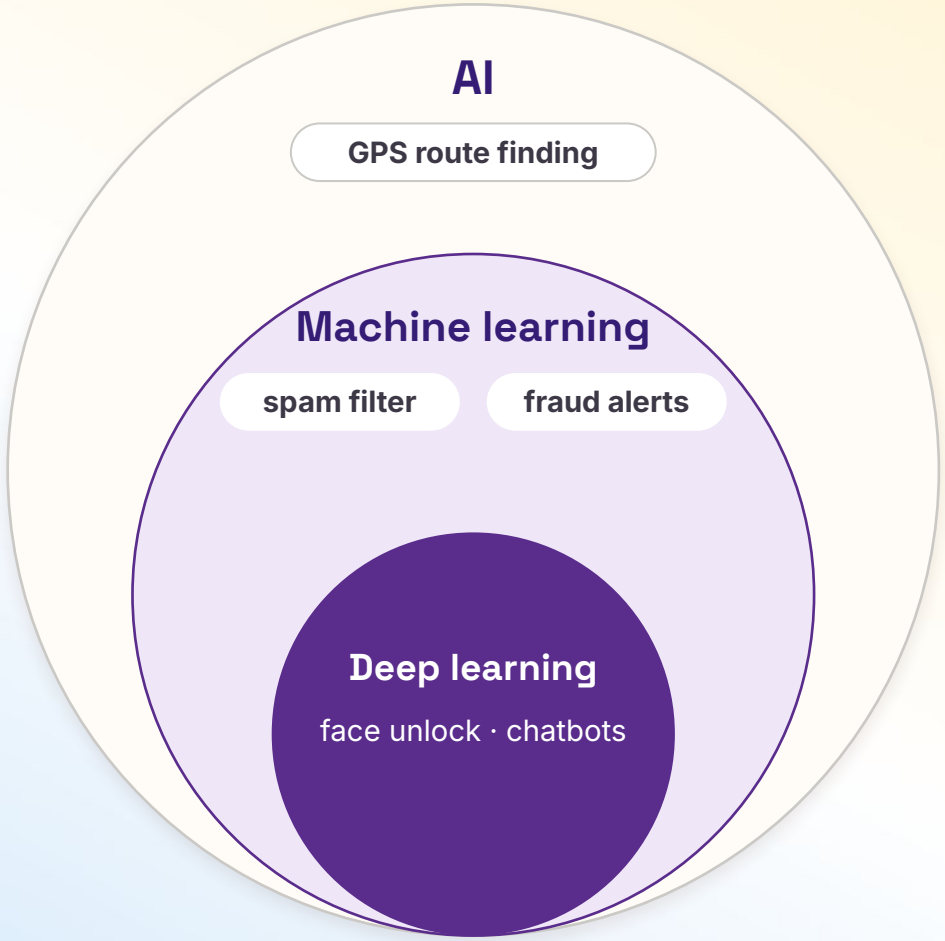
datasciencebootcamp@nyu.edu for questions, extensions and attendance approvals.

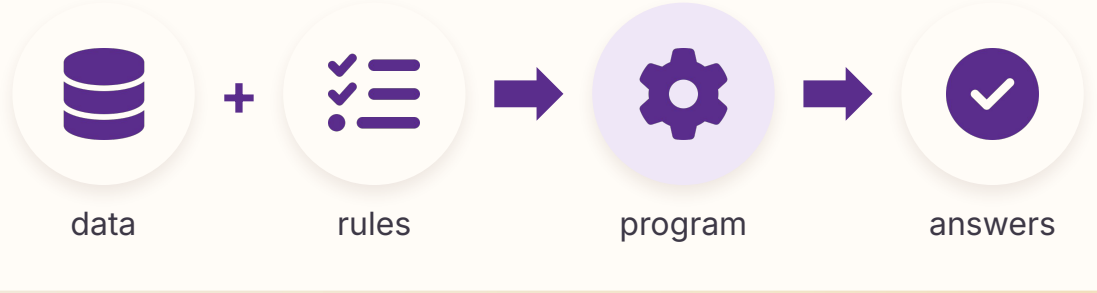
1:1 coaching

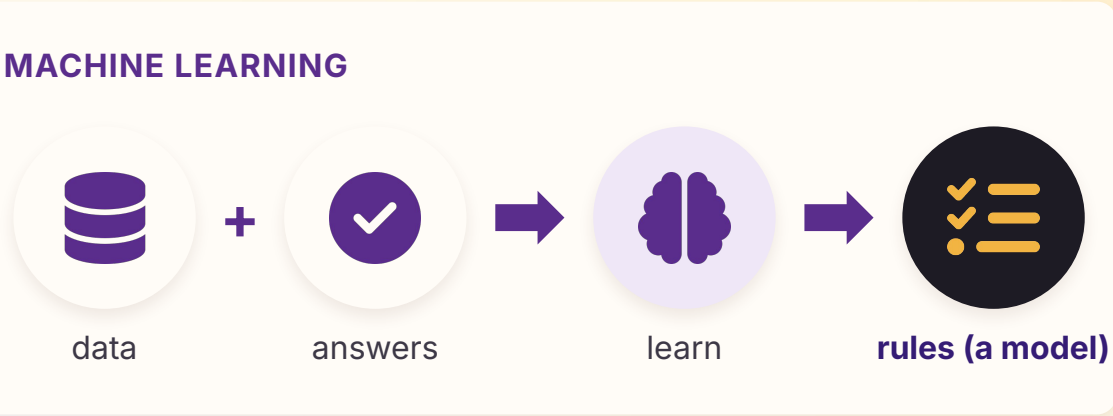
Optional 30-minute technical interview prep. Book it through Handshake.

SECTION 1 · AI VS. ML VS. DEEP LEARNING

ML is the part of AI that learns rules from data



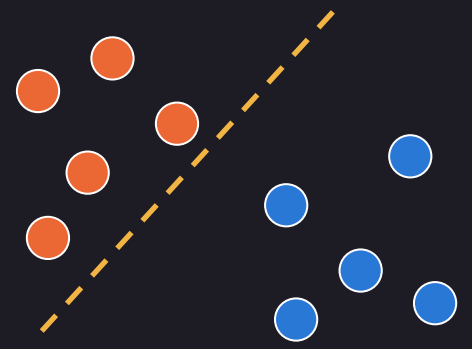




SECTION 2 · TYPES OF MACHINE LEARNING

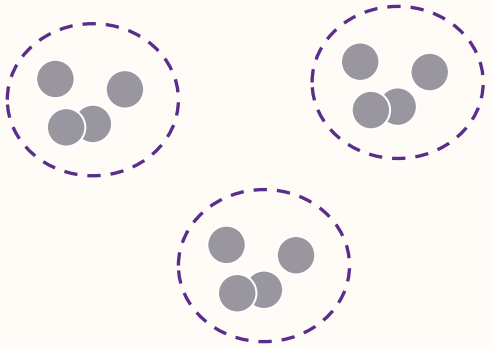
Does the data come with answers?

SUPERVISED



answers given: learn to predict them

UNSUPERVISED



no answers: find the groups

REINFORCEMENT



no answers: learn from rewards

COLAB SAMPLE DATA · FIND THE ANSWER COLUMN

California Housing

longitude	latitude	rooms	population	median_income	**median_house_value**

answer = **a number**

MNIST digits

label	px0	px1	px2	...	px783

answer = **a category (0-9)**

SECTION 3 · KEY ML TERMINOLOGY

Every term, pointed at one table

FEATURES X | LABEL y | SPLIT

	field_of_study	starting salary	ai_exposure	salary 4 yr	set
EXAMPLE →	**Computer Science**	$70,906	1.17	$100,385	train
	Mechanical Engineering	$70,705	1.18	$91,129	test
	Economics	$51,815	1.29	$81,228	train
	Psychology, General	$31,680	1.26	$50,152	test
	Dance	$21,878	-1.16	$39,716	train

FEATURES X

$70,906

starting salary

→ **MODEL**

`1.07 × x + 17,534`

parameters: slope, intercept

→ **PREDICTION y_hat**

$93,395

the model's answer

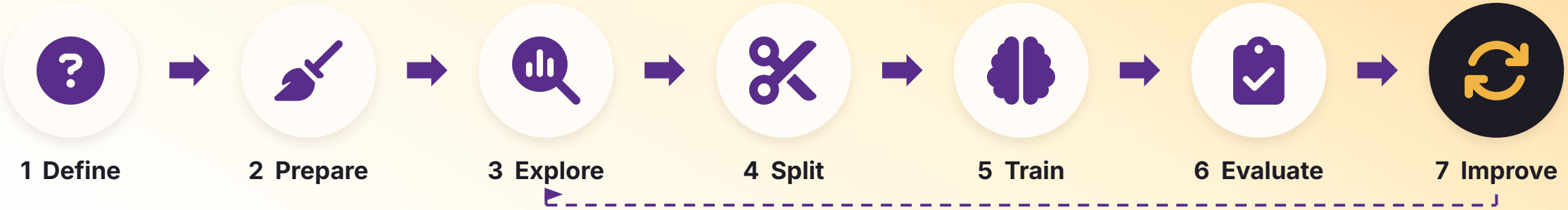
→ **TRUE LABEL y**

$100,385

miss: $6,990

SECTION 4 · THE MACHINE LEARNING WORKFLOW

Seven steps, and the last one loops back



QUESTION 1 · REGRESSION

$ → $

starting salary → salary 4 years later

QUESTION 2 · CLASSIFICATION



Who earns more: at least $80,000 after 4 years?

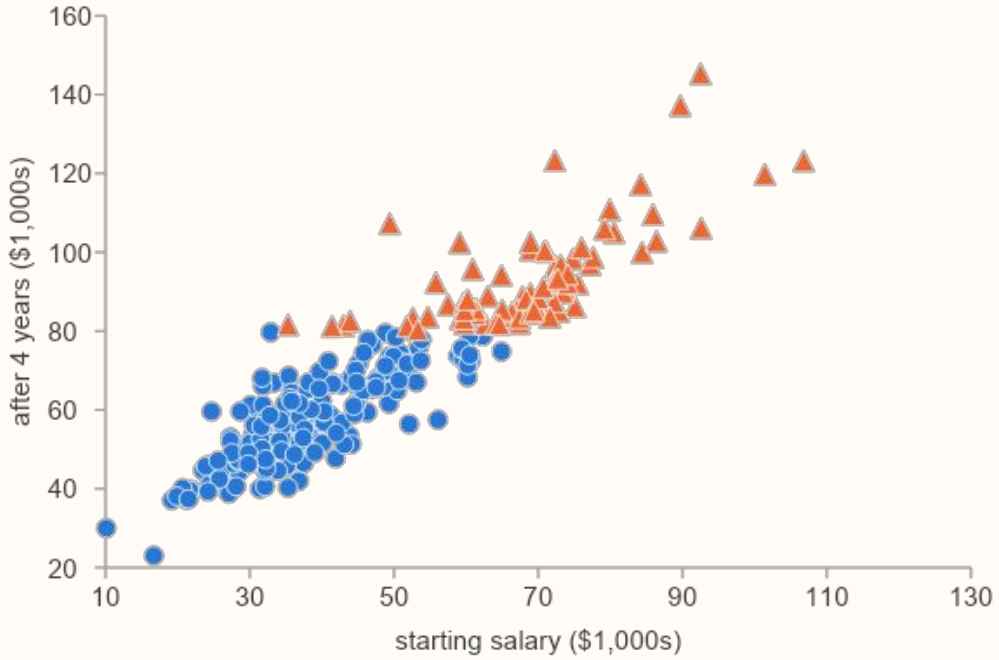
SECTION 4 · DATA STORY 1

AI exposure does not set the paycheck

STARTING SALARY vs. SALARY AFTER 4 YEARS

r = 0.92

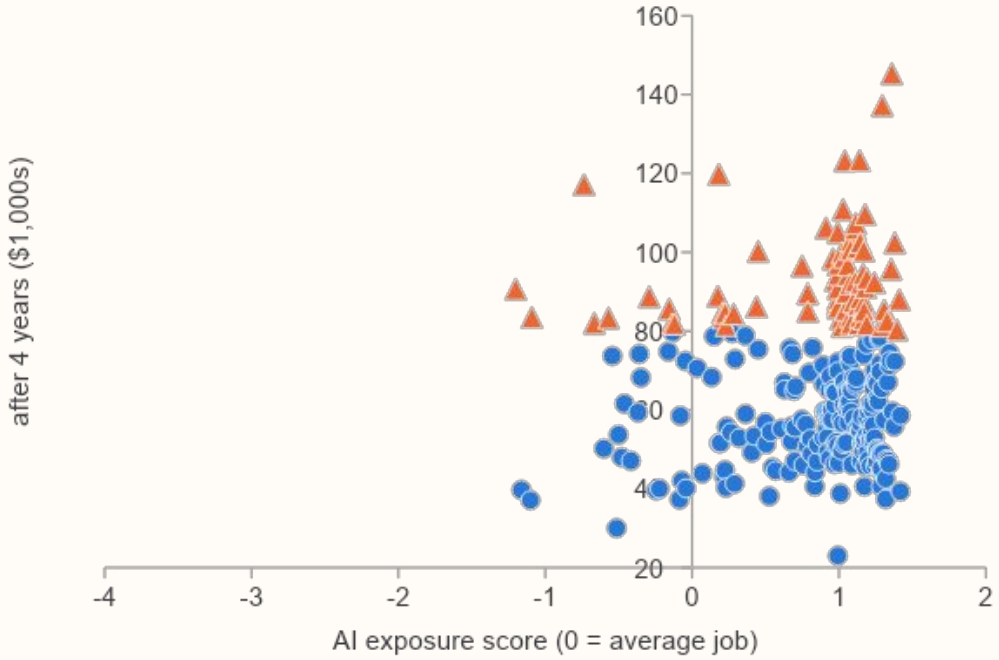
a strong feature



AI EXPOSURE vs. SALARY AFTER 4 YEARS

r = 0.08

almost no signal

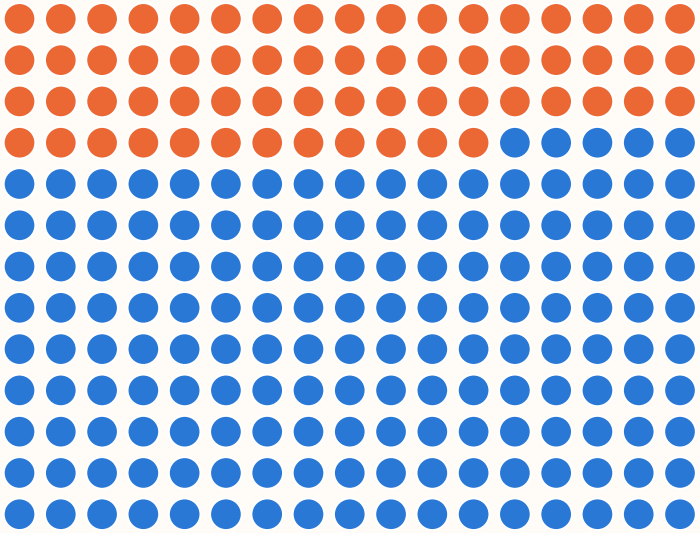


● high earner ● below $80k

SECTION 5 · TRAIN-TEST SPLIT

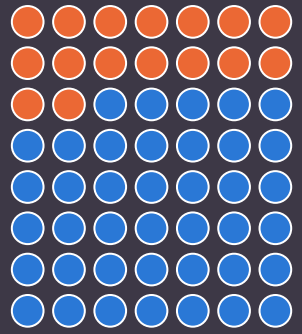
Hide 20% of the majors until the very end

TRAINING · 221 MAJORS



28.5% high earners

TEST · 56



28.6% high

Practice questions

training set: learn from these

Final exam, new questions

test set: checked once

No peeking

seeing the test early = leakage

```
test_size=0.2   random_state=42   stratify=high_earner
```

SECTION 6 · REGRESSION VS. CLASSIFICATION

Post Training: Predict a number or a category

REGRESSION · THE LABEL IS A NUMBER



Average miss **$5,564** vs. always-guess-the-average $15,959

CLASSIFICATION · THE LABEL IS A CATEGORY

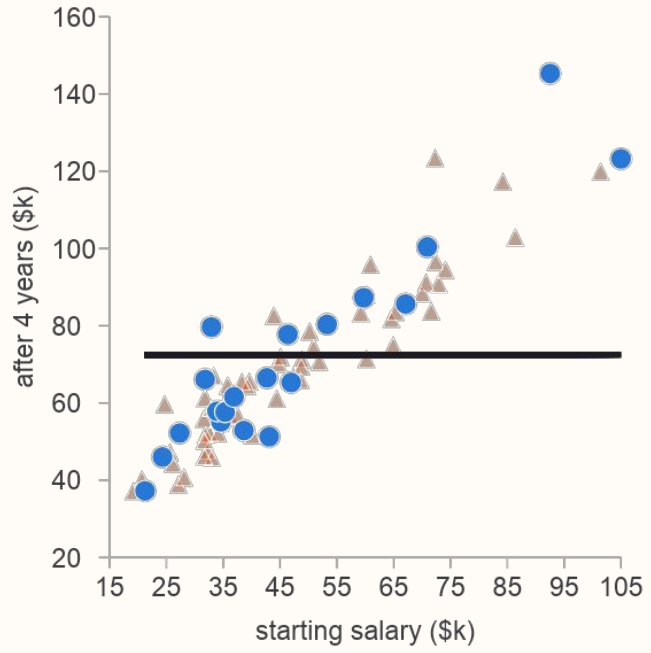


Rule **start > $55k → high earner**

SECTION 7 · OVERFITTING AND UNDERFITTING

Too simple, just right, too wiggly

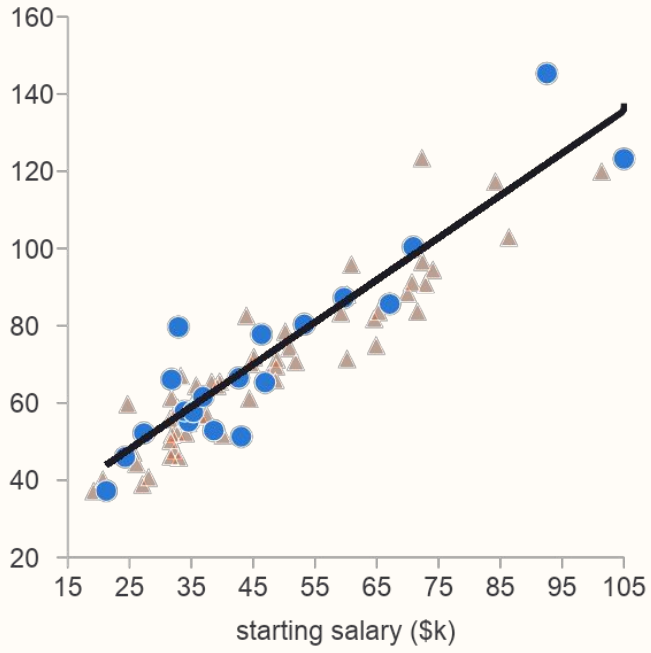
UNDERFIT · DEGREE 0



20.0 train MAE

16.9 test MAE

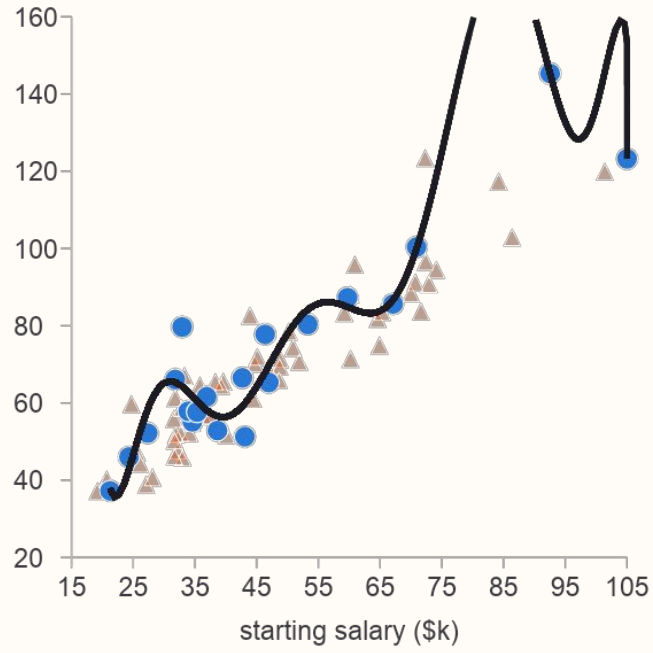
GOOD FIT · DEGREE 1



7.0 train MAE

6.7 test MAE

OVERFIT · DEGREE 10



4.3 train MAE

11.8 test MAE

SECTION 8 · CONFUSION MATRIX, PRECISION, RECALL

Sort every test major into four boxes

	RULE SAYS NO	RULE SAYS YES
truly no	38 true negatives	2 false alarms
truly yes	1 miss	15 true positives

PRECISION of 17 it called yes, how many are right?



15 / 17 = 0.88

RECALL of 16 real yeses, how many found? (○ = missed)



15 / 16 = 0.94

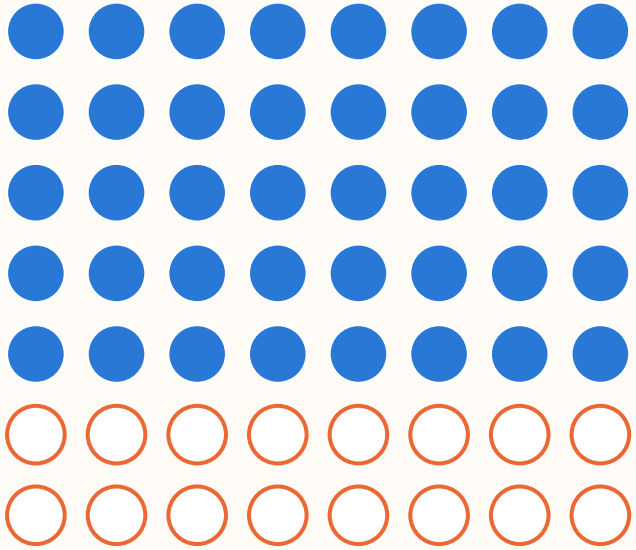
ACCURACY

53 / 56 = 0.95

SECTION 8 · DATA STORY 2

Accuracy can lie

LAZY MODEL · ALWAYS SAYS NO



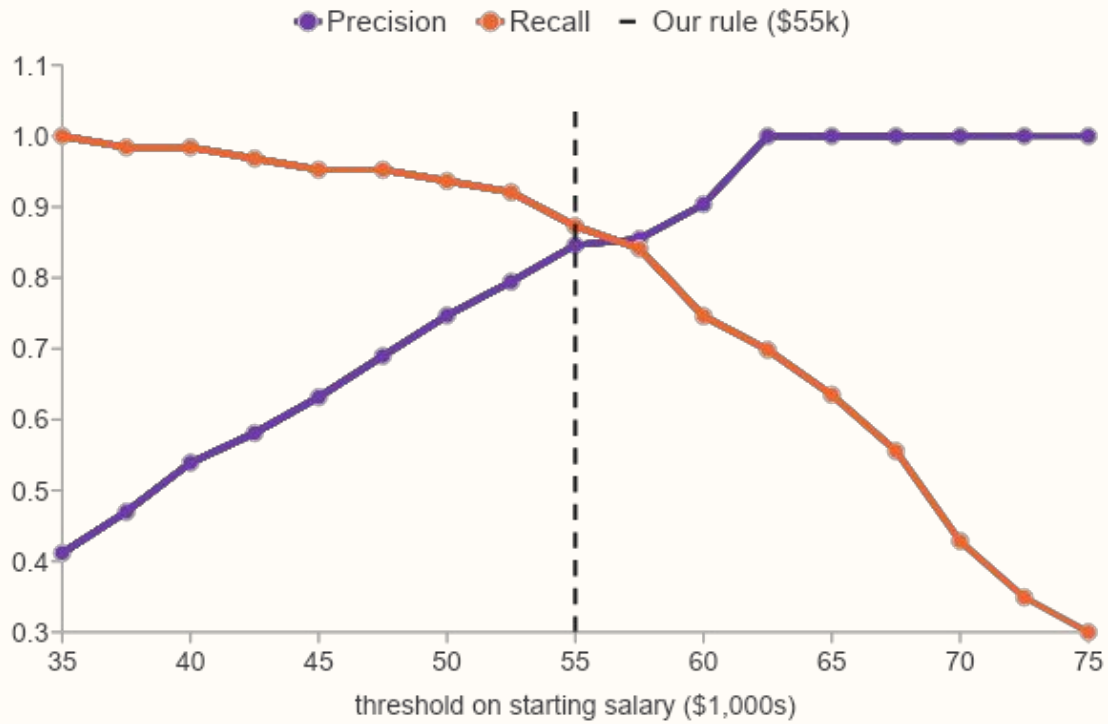
71%

accuracy: 40 right

0%

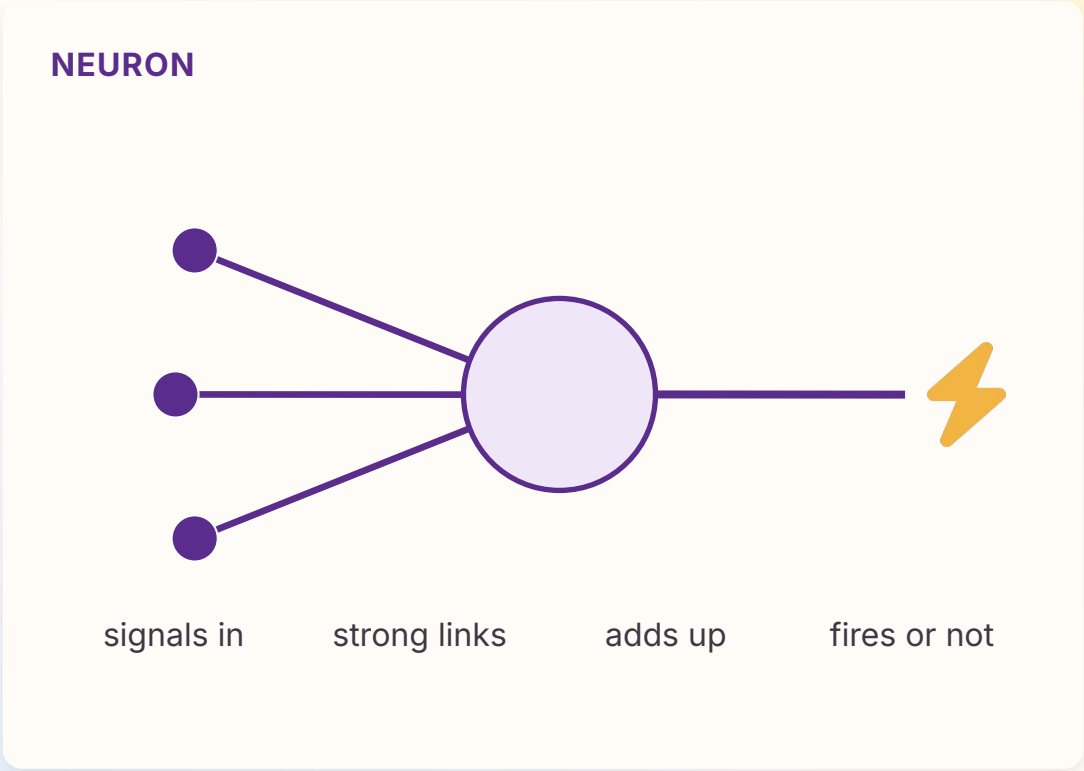
recall: 0 of 16 found

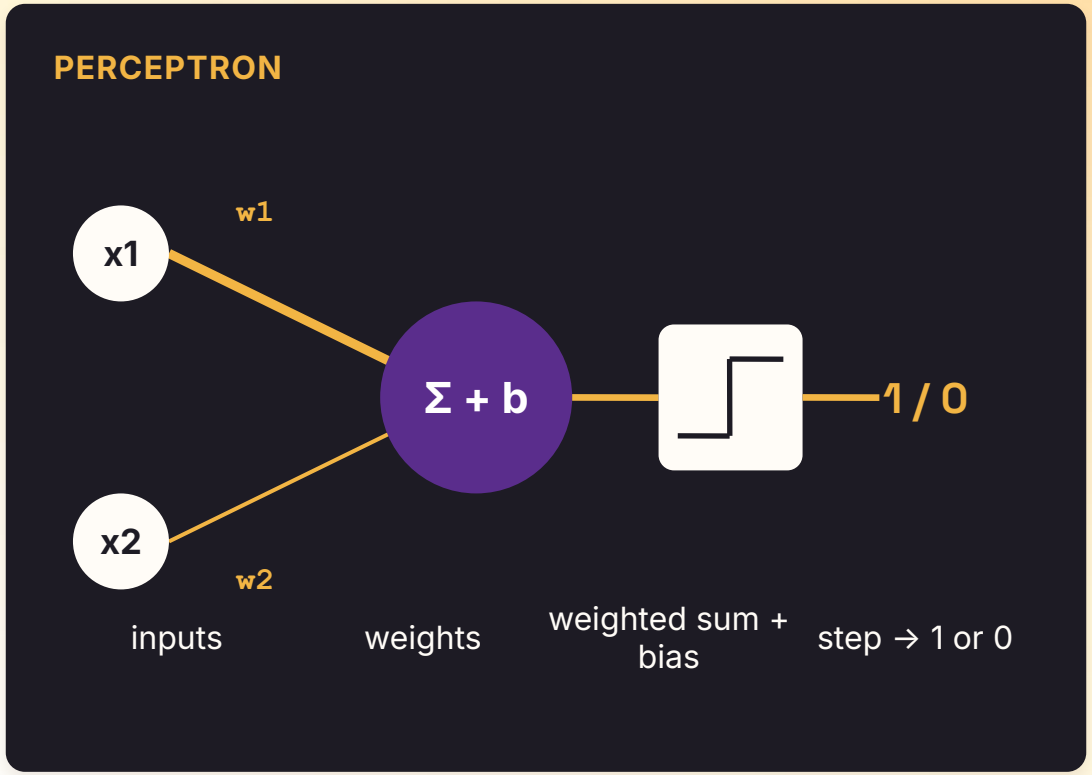
CHOOSING THE THRESHOLD (TRAINING SET)



SECTION 9 · INTRODUCTION TO THE PERCEPTRON

Neuron vs Perceptron: that says yes or no





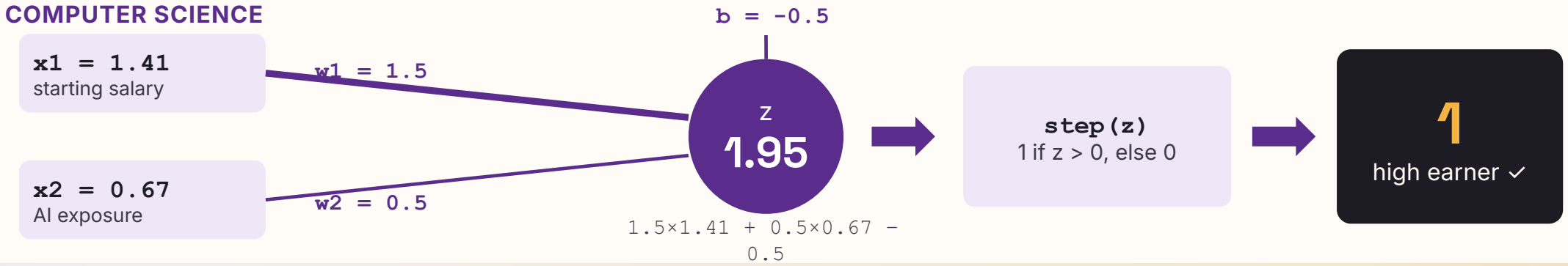
Basic Neuron - Building block of nerve.

Basic Perceptron - Building block of neural net.

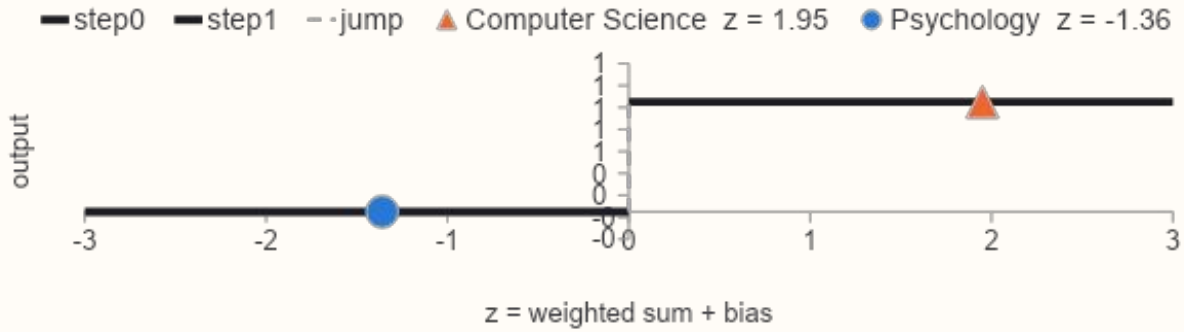
SECTION 10 · INPUTS, WEIGHTS, BIAS, ACTIVATION

Run one major through the perceptron

COMPUTER SCIENCE



ACTIVATION · THE STEP FUNCTION



SCALE FIRST · STANDARDIZE

starting salary

AI exposure

both, scaled

0 = average for every feature

SECTION 11 · TRAINING A PERCEPTRON

Move the line after every mistake

LEARNED BOUNDARY · TRAINING MAJORS, SCALED



0

correct

nothing changes

+1

miss

line moves toward it

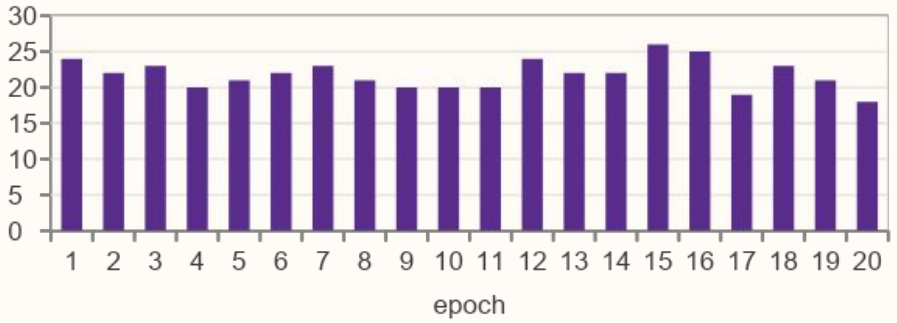
−1

false alarm

line moves away

MISTAKES PER EPOCH

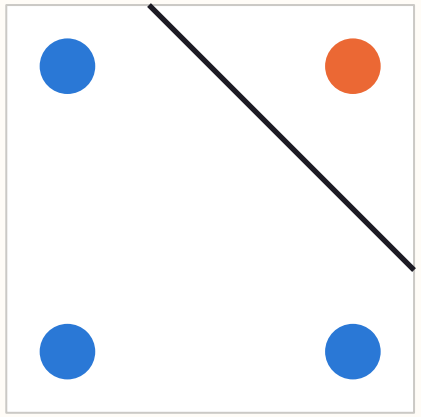
never reaches 0: the classes overlap



SECTION 12 · LINEAR SEPARABILITY AND XOR

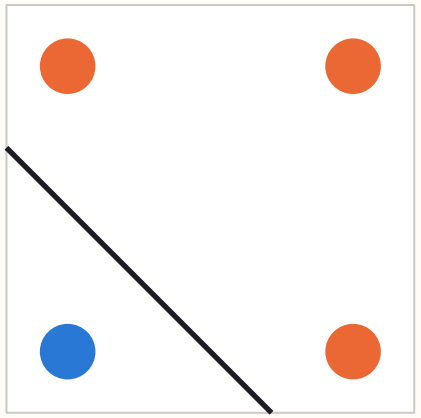
One straight line cannot solve XOR

AND



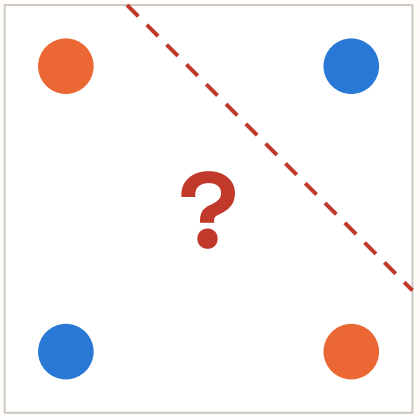
1 line ✓

OR



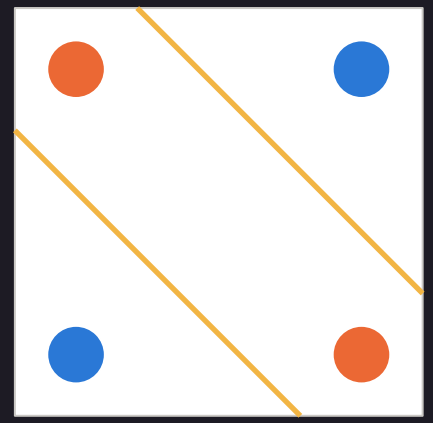
1 line ✓

XOR



1 line ✗

XOR, 2 LAYERS



2 lines ✓

● output 1 ● output 0

SECTION 13 · FROM SCRATCH AND WITH SCIKIT-LEARN

Perceptron From Scratch Code vs Library

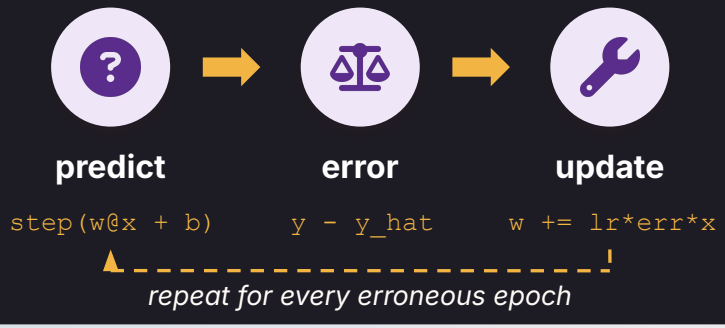
FROM SCRATCH

```
y_hat = 1 if w@x + b > 0 else 0
w += lr * (y - y_hat) * x
b += lr * (y - y_hat)
```

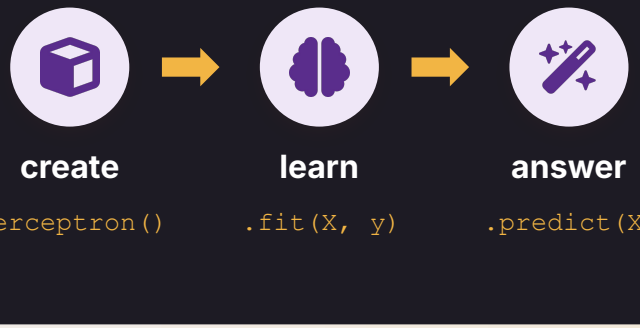
SCIKIT-LEARN

```
p = Perceptron(eta0=0.1, max_iter=20)
p.fit(X_train_s, y_train_c)
p.predict(X_test_s)
```

FROM SCRATCH · A LOOP AND ONE RULE



SCIKIT-LEARN · THREE CALLS



SUMMARY

Ten ideas from today



ML

learns rules from data



Supervised

data comes with answers



X and y

features in, label out



Split

judge on unseen data



Learning rule

fix it when wrong



Regression

number vs. category



Classification

predict a category



Perceptron

weights, bias, step



Overfitting

great train, poor test



Evaluation

accuracy can lie